

SCT221-C004-0755/2017

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SMA2101 : Calculus 1

1. a $f(x) = \sqrt{4-x^2} = \sqrt{-1(x^2-4)} = \sqrt{-1(x-2)(x+2)} \neq \sqrt{0}$
 $-1(x-2)(x+2) \neq 0 \therefore -2 \leq x \leq 2$ (x Between -2 and 2)
 Domain = $(-2, 2)$ Range = $(0, 2)$

b $\lim_{x \rightarrow 2} \frac{(x^2-4)}{x-2} = \lim_{x \rightarrow 2} \frac{((x-2)(x+2))}{x-2} = \lim_{x \rightarrow 2} x+2$
 $\Rightarrow 2+2 = 4$

c $y = (3x^2+2x)(x-5) = 3x^3 + 2x^2 - 15x^2 - 10x = 3x^3 - 13x^2 - 10x$
 $dy/dx = 9x^2 - 26x - 10$

d curve $y = x^3 - 4x$ at $x = 2$

Point $\Rightarrow \lim_{x \rightarrow 2} x^3 - 4x \Rightarrow (2)^3 - 4(2) \Rightarrow 0 \therefore$ points $(2, 0)$

m @ gradient = $dy/dx = 3x^2 - 4$ but $x = 2$ $m = 3(2)^2 - 4 = 8$

but $y = mx + c \Rightarrow y = 8x + c$

but $c = dy - dx \Rightarrow (y-0) - (x-2) \Rightarrow c = y - x + 2$

$c = y - 8x$ or $y - x + 2$

but $y = mx \Rightarrow (y-0) = 8(x-2) \rightarrow$ tangent $\Rightarrow y = 8x - 16$

2a $s(t) = t^3 - 6t^2 + 9t$

$ds/dt = 3t^2 - 12t + 9 \Rightarrow$ velocity

$ds/dt = 6t - 12 \Rightarrow$ acceleration

distance at $t=0$ to $t=3$

but velocity = $ds/dt = 3t^2 - 12t + 9$

~~distance = $\int ds = \int (3t^2 - 12t + 9) dt = (3t^3 - 12t^2 + 9t) = (3 - 12 + 9) = 0$~~

$V(t) = 3t^2 - 12t + 9$

$p \rightarrow 27 \quad s \rightarrow -12 \quad (-9, -3)$

$= 3t^2 - 9t - 3t + 9 = 3t^2 - 12t + 9$

$= 3t(t-1) - 9(t-1) = (3t-9)(t-1)$

when $V=0 \Rightarrow (3t-9)(t-1)=0 \quad t=1$ or 3

But curve position $\rightarrow t^3 - 6t^2 + 9t \therefore$ position 1 = $1^3 - 6(1)^2 + 9(1) = 4$
 position 2 = $3^3 - 6(3)^2 + 9(3) = 0$

Distance = Δ position = $4 - 0 = 4$

Total distance = $4 \times 2 = 8 \Rightarrow$ absolute

b rectangle (L, w)



$$\text{area} = L \times w = 1200 \text{ m}^2$$

$$\text{Perimeter} = 2(L + w) = 2L + 2w$$

$$\text{but area} = L \times w = 1200 \quad \therefore L = 1200/w$$

$$\text{replace} \quad \text{Perimeter} = 2\left(\frac{1200}{w}\right) + 2w = \frac{2400 + 2w^2}{w}$$

$$\text{minimum perimeter} \Rightarrow \text{perimeter} \rightarrow 0$$

$$P_{\rightarrow 0} = \frac{2400 + 2w^2}{w} \neq 0 \Rightarrow 2w^2 = -2400 \Rightarrow w^2 = 1200$$

$$w = \sqrt{1200} \quad \therefore L = \frac{1200}{\sqrt{1200}}$$

$$w \approx 34.64 \text{ m}$$

$$L \approx 34.64 \text{ m}$$

$$\text{dimensions} \Rightarrow \underline{\underline{34.64 \text{ m by } 34.64 \text{ m}}}$$